Who’s that bot? A Pokemon Showdown Bot Through Reinforcement Learning

Abstract

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Related Works

Winning at Pokémon Random Battles Using Reinforcement Learning

Wang (2024) presents a bot designed to play random battles in Generation 4. The agent is modeled as a Partially Observable Markov Decision Process (POMDP), trained using Proximal Policy Optimization (PPO), and further improved at inference time through Monte Carlo Tree Search (MCTS).

Introduction

Problem Statement

Since Deep Blue defeated Kasparov in 1997, the use of AI in turn-based games has been extensively explored. Games like chess and Go now have AIs capable of defeating the best human players, Stockfish in chess and AlphaGo in Go, yet there is still no bot capable of consistently beating the top Pokémon VGC players to this day. And it's not that Pokémon is necessarily harder than chess, but two things are certainly true. First, it has a larger number of possible game states than chess. Second, it's much harder to tell whether an AI is winning or losing at any given moment, unlike chess, where having more pieces and more paths to checkmate are reliable indicators of advantage. In Pokémon, you can be down two Pokémon and still be winning.

Background

Pokemon VGC

Pokémon VGC is Nintendo's official format for Pokémon battles. It consists of 2-on-2 matches, where each player brings a team of 6 Pokémon, but at the start of each battle they select only 4 of them, and of those 4 only 2 can be on the field at the same time. Each Pokémon can hold an equipped item, and throughout the battle players take turns simultaneously choosing an action for each of their two active Pokémon, either attacking or switching in another member of the team. The winner is whoever leaves the opponent with no Pokémon remaining, or whoever has more Pokémon standing when the battle timer runs out.

Pokemon Showdown

Pokémon Showdown is a free, open-source Pokémon battle simulator that runs directly in the browser. Unlike the official video games, where players need to invest hundreds of hours obtaining and training their Pokémon, Showdown allows anyone to build a competitive team

in a matter of minutes and start battling right away. This makes it the perfect tool for the competitive community to practice, test strategies, and prepare for official tournaments.And in this case train our IA.

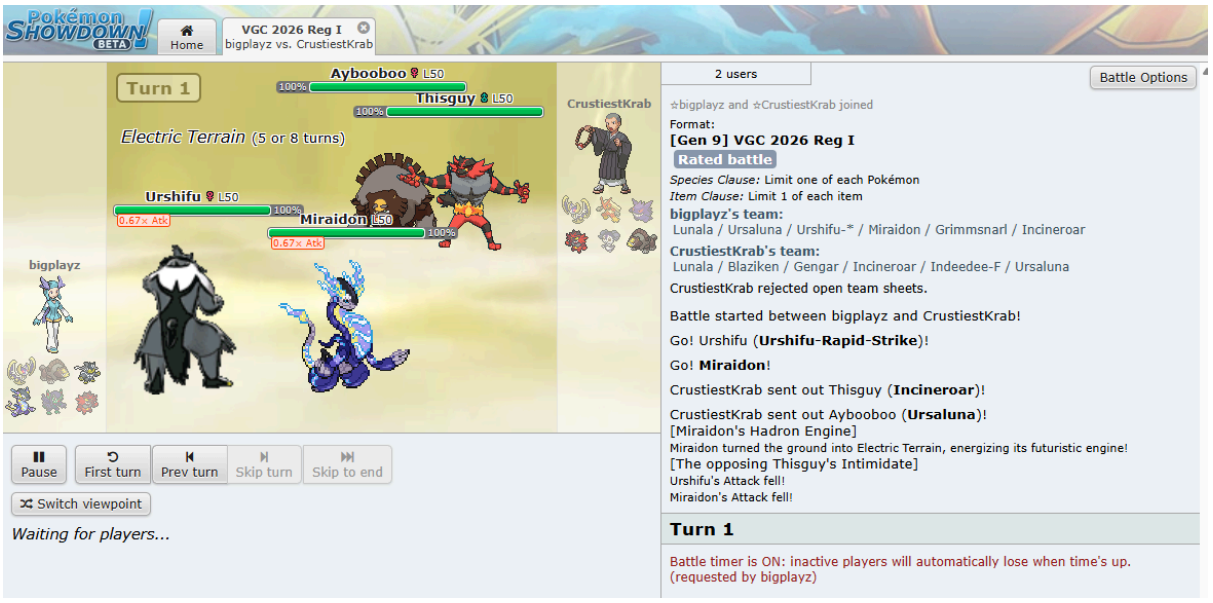

Figure 1.1: A Gen 9 VGC 2026 regulation I match in Pokemon Showdown.

Environment

Environment Description

The agent will be trained in a local Pokémon Showdown server running on Node.js, interacting with it through the Python library poke-env. Battles will take place in the VGC 2026 Regulation I format, and the agent will use a fixed team built by a human, which will remain the same throughout the entire training process. A battle ends when all Pokémon on one side have fainted.

State Space

At the start of each turn, the agent receives an almost complete state of the battle state. It knows the current HP of all its Pokémon, which Pokémon are still available in reserve, the moves, their held items, and any active stat boosts. On the opponent's side, since VGC 2026 Regulation I is played in an open sheet format meaning both teams are revealed at the start of the battle, the agent has full knowledge of the rival's current HP, the Pokémon visible on the field, their held items, their used moves, and any active stat boosts, the agent also receives information about the general battle state, including the current weather and terrain, any field effects such as Tailwind or Trick Room, which Pokémon on either side have fainted, and the current turn number.

State Space:
Own side**:**

- Current HP of all its Pokémon
- Which Pokémon are still available in reserve
- Available moves
- Held items
- Active stat boosts

Opponent's side:
- Current HP of all opponent's Pokémon
- Pokémon visible on the field
- Held items
- Moves that have been used
- Active stat boosts

General battle state:
- Current weather and terrain
- Active field effects
- Current turn number

Action Space

Each turn, the agent chooses an action for each of its two active Pokémon. For each one, it can either use one of its available moves or switch it out for a Pokémon from the reserve. When using a move, the agent must also specify a target if the move requires one, either one of the two opposing Pokémon or its own ally, whereas moves that hit all Pokémon on the field do not require a target selection. Additionally, once per battle the agent may Terastallize one of its active Pokémon, which changes its type and can be combined with a move action on the same turn.

Action Space:
- Use a move
- Switch out
- Terastallize

Reward Function

The reward function is based on Wang (2024). The agent receives a reward of +1 upon winning a battle, -1 upon losing, and 0 on every other turn. This reward structure allows the agent to learn strategies that play out over multiple turns, such as moves with effects that last several turns or setups that only pay off later in the battle, something that would not be possible if the reward function assigned value to individual turn outcomes.

Methods

Training

The agent will play against itself to train the model. We will save the SimpleHeuristicsPlayer, a pokemon bot, for validation. According to Wang (2024), the bot is easy enough for the agent to beat, but difficult enough for it to learn from. This balance is precisely why RandomPlayer and MaxBasePowerPlayer were avoided, RandomPlayer selects actions at random, meaning the agent would learn nothing meaningful from it, while the MaxBasePowerPlayer always picks the highest base power move, which could teach the agent the bad habit of always assuming that the strongest move is the best choice.

Validation

The metrics used to measure the model's success will be based on Wang (2024). We will use win rate, against SimpleHeuristicsPlayer, as the primary success metric, measured every 20,000 steps, which according to the paper corresponds to approximately 200 battles against the bot.

Agent

Model 1

Model 2

Results

Results and metrics

Analysis and comparison

Conclusions

Resume

Limitations

Future Possible Work

References

[1] Wang, J. (2024). Winning at Pokémon random battles using reinforcement learning (Master's thesis). Massachusetts Institute of Technology.